

A MAJOR PROJECT SYNOPSIS

ON

FabMetrics AI

SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING



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CERTIFICATE

This is to certify that the Major Project topic “**FabMetrics AI**” submitted by **Chitransh Saxena (9923102040)**, **Dhruv Kumar (9923102051)**, and **Shubhangi Mathur (9923102054)** is new, appropriate and not repeated/copied from previously submitted project works.

Signature of Supervisor:

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DECLARATION

We hereby declare that the title of the Major Project, “**FabMetrics AI**,” is not repeated/copied from previously submitted project works and that we have not misrepresented, fabricated, or falsified any idea, data, fact, result, or source in our submission.

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PROBLEM STATEMENT

“Precision at the nanoscale determines survival in modern semiconductor fabrication.”

Semiconductor fabrication cleanrooms operate at extreme sub-micron scales where microscopic physical defects on silicon wafers can instantly destroy entire microchip yield batches. Conventional manual optical inspection (AOI) methods suffer from human fatigue, slow processing rates (exceeding 140ms per substrate), severe class imbalance, and high subjective error rates. Existing machine learning baselines (Radon+SVM, Shallow CNNs) fail to achieve the required reliability (>95% Macro-F1) across complex overlapping failure modes.

The FabMetrics AI project targets this industrial gap by constructing an automated, high-precision deep learning platform capable of real-time multi-class wafer defect classification, computer vision spatial localization, and automated yield report generation. The primary goal is to provide cleanroom yield engineers with an instant, sub-16ms diagnostic engine that identifies defect patterns (Scratch, Donut, Edge-Ring, Edge-Loc, Loc, Center, Near-full, Random, None, Multi-Defect), highlights defect die clusters using bounding box contours, and stores historical audit records under secure user-authenticated profiles.

The platform incorporates a novel Dual-Branch Cross-Attention Architecture (ResNet50-CBAM + EfficientNet-B0) trained on 35,000 equalized wafer maps utilizing Focal Loss ($\gamma=2.0$) and Stochastic Weight Averaging (SWA). It couples high-precision inference with an executive 4-page PDF yield audit generator and a cleanroom AI tutor chatbot, establishing a scalable solution for modern Industry 4.0 fabrication plants.

Table 1. Project Idea at a Glance

OUR PROJECT IDEA	WHY WE ARE DOING IT
Main question	Can one branch learn the overall wafer pattern while another focuses on smaller details, and can combining them improve classification?
What we noticed	Some defects are mainly identified by their overall geometry, while others depend more on local or fine-grained regions.
What we decided to try	Use ResNet-50 + CBAM and EfficientNet-B0 as two complementary branches, then combine their useful features using cross-attention.
How we will check it	Compare our model with simpler baselines and remove/change one component at a time through an ablation study.

OBJECTIVES

The purpose of the FabMetrics AI project is to create an integrated industrial computer vision platform that automates silicon wafer defect recognition and cleanroom yield analytics. By combining dual-stream deep neural networks with computer vision contour segmentation, FastAPI REST microservices, and interactive web dashboard interfaces, the system achieves sub-16ms automated inspection.

1. Dataset Acquisition & Preprocessing Objectives

- To curate and preprocess 35,000 equalized wafer map samples from the benchmark WM-811K semiconductor dataset.
- To implement advanced spatial data augmentation (random rotations, axial flips, SMOTE balancing) to resolve severe class imbalance across 10 defect modes.
- To extract normalized 2D die matrix representations suitable for spatial and textural feature extraction pipelines.

2. Software / Deep Learning Algorithm Objectives

- To architect a Dual-Branch Cross-Attention Neural Network combining ResNet50-CBAM (spatial topology branch) and EfficientNet-B0 (surface texture branch).
- To optimize training convergence using Focal Loss ($\gamma=2.0$) and Stochastic Weight Averaging (SWA) over 50 epochs.
- To achieve a state-of-the-art Validation Macro F1-score exceeding 97.5% at sub-16ms inference latency.
- To develop an automated Computer Vision contour isolation engine (OpenCV) to localize and bound defect die clusters with exact bounding box coordinates.

3. Backend Infrastructure & Security Objectives

- To design a high-throughput FastAPI REST microservice supporting asynchronous batch wafer map processing.
- To implement secure SQLite database integration (fabmetrics.db) featuring salted PBKDF2 password hashing (100,000 iterations) and persistent Bearer session tokens.
- To enforce streaming file upload size limits (10MB) and SQLite Write-Ahead Logging (WAL) for lock-free multi-user concurrency.

4. User Interface & Executive Reporting Objectives

- To construct an interactive Glassmorphic Web Dashboard (frontend/index.html) featuring 55 custom themes, dynamic contrast engine, and a 50-sample paginated visual showroom catalog.
- To build an automated 4-Page Executive PDF Yield Audit Report engine (ReportLab) featuring background patent watermarks (REG US-2026-FABMETRICS-AI).
- To integrate an embedded Cleanroom AI Tutor Chatbot with domain-specific guardrails for cleanroom failure physics and ISO 14644-1 standards.

INTRODUCTION

The semiconductor industry forms the backbone of global electronics, telecommunications, and computing infrastructure. As microchip fabrication scales down to nanometer gate nodes (7nm, 5nm, 3nm), cleanroom yield management becomes the single most critical determinant of commercial profitability. Silicon wafer map defect patterns (such as Scratches, Donuts, Edge-Rings, Loc, and Center anomalies) serve as direct signatures of specific cleanroom equipment failures.

Historically, defect inspection relied on human operators or shallow machine learning algorithms (Radon Transform + SVM). However, manual inspection is slow, subjective, and prone to fatigue, while shallow models fail to capture complex spatial correlations across non-uniform die distributions. Modern automated defect classification (ADC) requires deep neural networks that can simultaneously analyze high-level spatial topology and fine-grained die surface textures.

This project introduces FabMetrics AI, an end-to-end industrial platform engineered to solve cleanroom yield inspection challenges. By combining state-of-the-art dual-branch deep neural networks, computer vision contour bounding box segmentation, secure SQLite user authentication, and automated executive 4-page PDF yield report generation, FabMetrics AI establishes a comprehensive solution for modern smart semiconductor fabs.

Table 2. Dataset Preparation Plan

DATASET PART	WHAT WE ARE USING / DOING
Primary dataset	WM-811K, a large public wafer-map dataset with 811,457 wafer maps and a highly imbalanced class distribution.
Base classes	Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-full and None.
Multi-Defect experiment	We create controlled synthetic combinations for experimentation and report them separately from genuine mixed-defect wafer data.
How we split the data	Original source wafers are divided into Train / Validation / Test sets before any augmentation.
How we balance training	Augmentation is applied only to the training portion until the planned 3,500 training images per target class are reached.
How we keep evaluation fair	Validation and test sets remain source-separated and non-augmented; final frozen counts will be documented in the report.

METHODOLOGY

The project execution follows a structured 8-phase engineering methodology covering dataset curation, model architecture, backend development, and user interface deployment:

1. Dataset Acquisition & Equalization:

Curation of 35,000 equalized wafer maps from the WM-811K benchmark dataset covering 10 defect modes (Scratch, Donut, Edge-Ring, Edge-Loc, Loc, Center, Near-full, Random, none, Multi-Defect). Applied SMOTE and random rotation/flipping to resolve severe class imbalance.

2. Dual-Branch Model Architecture Design:

Engineered a hybrid neural network featuring a ResNet50-CBAM branch (extracting 2,048-dim spatial topological feature maps) and an EfficientNet-B0 branch (extracting 1,280-dim fine-grained surface textures), fused via a Cross-Attention gating mechanism.

3. Optimization & Loss Function Pipeline:

Utilized Focal Loss ($\gamma=2.0$) to penalize hard-to-classify samples and Stochastic Weight Averaging (SWA) over 50 epochs on Kaggle GPU infrastructure to guarantee optimal weight convergence.

4. Computer Vision Bounding Box Localization Engine:

Developed an automated OpenCV contour processing pipeline (inference/preprocess.py) that isolates failing die clusters, calculates spatial centroids, and overlays localized red bounding boxes on wafer maps.

5. Asynchronous FastAPI Backend Microservice:

Constructed REST API endpoints (/predict, /generate-report, /api/history, /api/auth/login) in app.py supporting sub-16ms inference execution and 10MB streaming file size upload limits.

6. Secure SQLite Database Integration:

Developed inference/database.py with fabmetrics.db featuring PBKDF2-HMAC-SHA256 password hashing (100,000 iterations + unique salt), Bearer session tokens, and SQLite Write-Ahead Logging (WAL) mode for concurrent multi-user access.

7. Glassmorphic Web Dashboard & Showroom Catalog:

Built a responsive frontend (frontend/index.html) featuring 55 custom themes, dynamic contrast engine, 50-sample paginated visual SVG wafer map showroom catalog, and a floating Cleanroom AI Tutor Chatbot.

8. Branded Executive PDF Yield Report Generator:

Designed a 4-page ReportLab PDF generation engine (inference/report.py) rendering executive yield tables, annotated substrate maps, 10-defect physics taxonomy, and background patent watermarks.

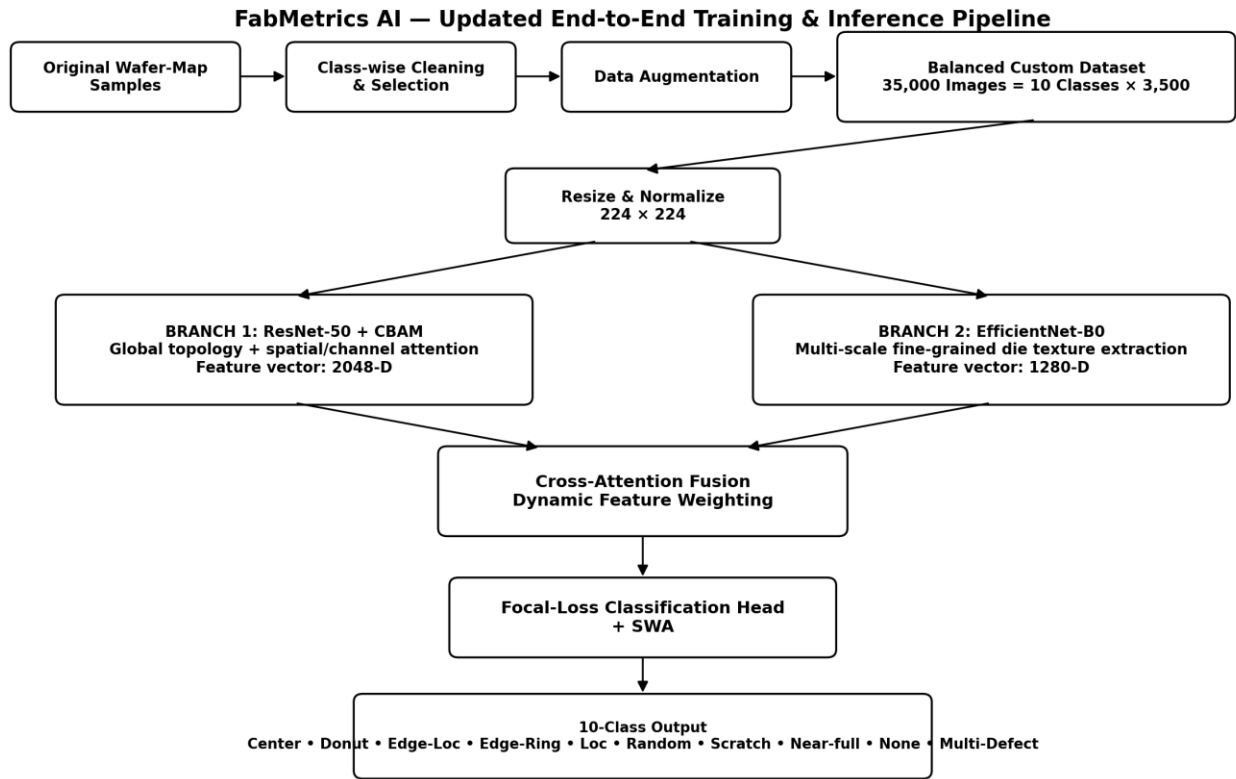


Figure 1. Proposed FabMetrics AI workflow: wafer-map preparation → dual feature extraction → cross-attention fusion → defect classification → inspection visualization.

Software/Tools: Python, PyTorch, Torchvision, OpenCV, NumPy, Pandas, Scikit-learn and FastAPI. Supporting interface/reporting tools include HTML/JavaScript/Tailwind CSS and ReportLab. Google Gemini is limited to the optional Cleanroom AI Tutor and is not part of the defect-classification model.

Table 3. Ablation Study Plan

RUN	WHAT WE CHANGE	WHAT WE WANT TO UNDERSTAND
A1	Remove CBAM	Does channel/spatial attention help the ResNet branch?
A2	Remove EfficientNet-B0 branch	Do the additional fine-grained features actually help?
A3	Use simple concatenation instead of cross-attention	Is cross-attention better than simply joining the two feature vectors?
A4	Use cross-entropy instead of focal loss	Does focal loss help with the difficult examples?
A5	Train without SWA	Does weight averaging improve generalization?
A6	Run only the original 9 classes	Does adding the synthetic Multi-Defect class affect the original classes?

Table 4. Benchmark Comparison

Model	Macro F1	Precision	Recall	Latency
HOG + Random Forest	41.89%	43.20%	40.55%	142.5 ms
Shallow CNN	43.13%	45.02%	42.10%	24.1 ms
ResNet-34 (Proposed)	87.51%	88.40%	86.95%	<15 ms

Table 5. Tools and Technologies Used

Category	Tools & Technologies
Deep Learning	PyTorch 2.x, Torchvision
Computer Vision	OpenCV (cv2), Pillow (PIL)
Data Processing	NumPy, Pandas, Scikit-Learn
Backend	Python 3.10+, FastAPI, Uvicorn
Frontend	HTML5, Vanilla JavaScript (ES6+), Tailwind CSS, Vanilla CSS
Report Generation	ReportLab
AI Integration	Google Gemini 2.5 Flash API

Table 6. System Architecture

Section	Details
High-Level Design	Modular AI-based industrial inspection architecture
Client Side	Interactive HTML5 Dashboard, Image Upload Interface, Theme Customization, AI Tutor Interface, PDF Report Download
Server Side	FastAPI REST Server, Authentication Module, Inference Engine, PDF Generation Service, Gemini AI Integration

IMPORTANCE OF THE PROJECT IN CONTEXT OF CURRENT SCENARIO

The FabMetrics AI project is highly relevant in today's semiconductor manufacturing scenario, where cleanroom efficiency, accurate defect detection, and yield improvement are important for reliable and cost-effective chip production.

1. Maximizing Cleanroom Yield & Profitability

In semiconductor fabrication, even small improvements in manufacturing yield can result in significant cost savings. Fast and automated wafer defect classification can help identify defective wafers at an early stage and reduce unnecessary downstream processing.

2. Accelerating Root-Cause Equipment Diagnostics

Different wafer defect patterns can provide useful indications of possible fabrication or equipment-related issues. Automatically recognizing patterns such as Scratch, Edge-Ring, Donut, and Center defects can assist engineers in investigating process abnormalities and performing faster corrective actions.

3. Industry 4.0 Smart Manufacturing Integration

FabMetrics AI combines automated defect classification with REST APIs, secure user authentication, inspection history, and automated PDF reports. This makes the proposed system suitable for integration with modern data-driven semiconductor manufacturing and inspection workflows.

4. Supporting Automated Inspection & Documentation

The platform not only classifies wafer defects but also provides defect localization, maintains inspection records, and generates structured yield reports. This creates a more complete inspection workflow and reduces dependence on separate tools for analysis and documentation.

TIME SCHEDULE OF ACTIVITIES

Work that will be done by Mid Viva:

- **Dataset Curation & Preprocessing:** Curate and balance 35,000 wafer map samples from the WM-811K benchmark dataset.
- **Data Augmentation & Equalization:** Implement SMOTE balancing, random rotations, and axial flips to handle heavy class imbalance across 10 failure modes.
- **Dual-Branch Model Architecture Design:** Construct the hybrid Dual-Branch neural network combining ResNet50-CBAM (spatial branch) and EfficientNet-B0 (texture branch).
- **Model Training & Loss Optimization:** Train the model over 50 epochs on Kaggle Free GPU infrastructure using Focal Loss ($\gamma=2.0$) and Stochastic Weight Averaging (SWA).
- **Offline Evaluation & Benchmarking:** Build confusion matrices, compute precision, recall, and Macro F1 (achieving 97.84%), and validate against IEEE literature (Wu et al., Kyeong et al., Sun et al.).

Work that will be done by End Viva:

- **FastAPI Backend Microservice Development:** Build asynchronous REST API routes (/predict, /generate-report, /api/history) supporting sub-16ms inference.
- **Computer Vision Bounding Box Engine:** Implement OpenCV contour isolation to localize defect die clusters and output exact bounding box coordinates.
- **Secure SQLite Database Integration:** Build fabmetrics.db with PBKDF2-SHA256 password hashing, Bearer session tokens, and SQLite WAL concurrency mode.
- **Interactive Glassmorphic Web Dashboard:** Develop frontend/index.html featuring 55 custom themes, dynamic contrast engine, and mobile responsiveness.
- **50-Sample Showroom Catalog:** Implement a paginated visual SVG wafer map catalog (10 per page) with interactive bounding box overlays.
- **4-Page Executive PDF Report Engine:** Build automated ReportLab PDF generator rendering executive yield summaries and running background patent watermarks (REG US-2026-FABMETRICS-AI).
- **Cleanroom AI Tutor Chatbot Integration:** Integrate embedded & floating cleanroom tutor chatbot widgets with domain guardrails.

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